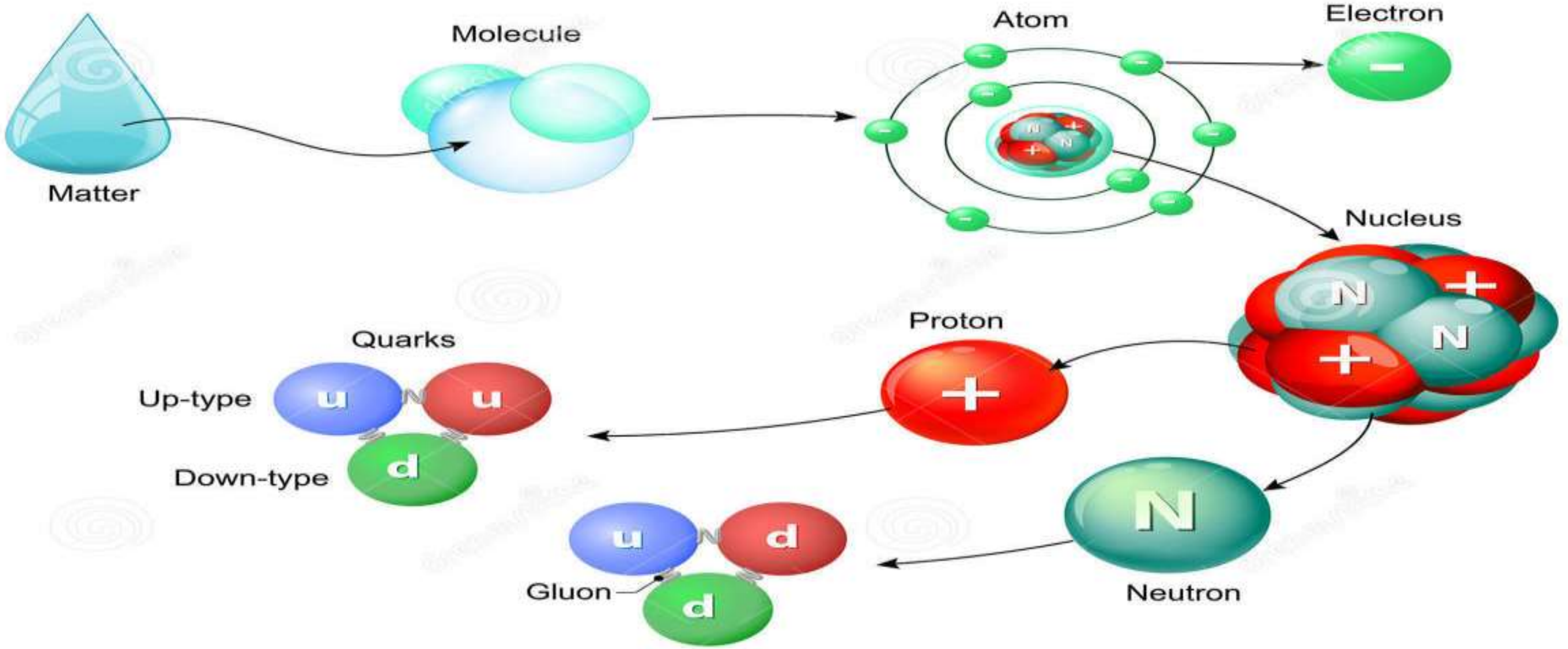
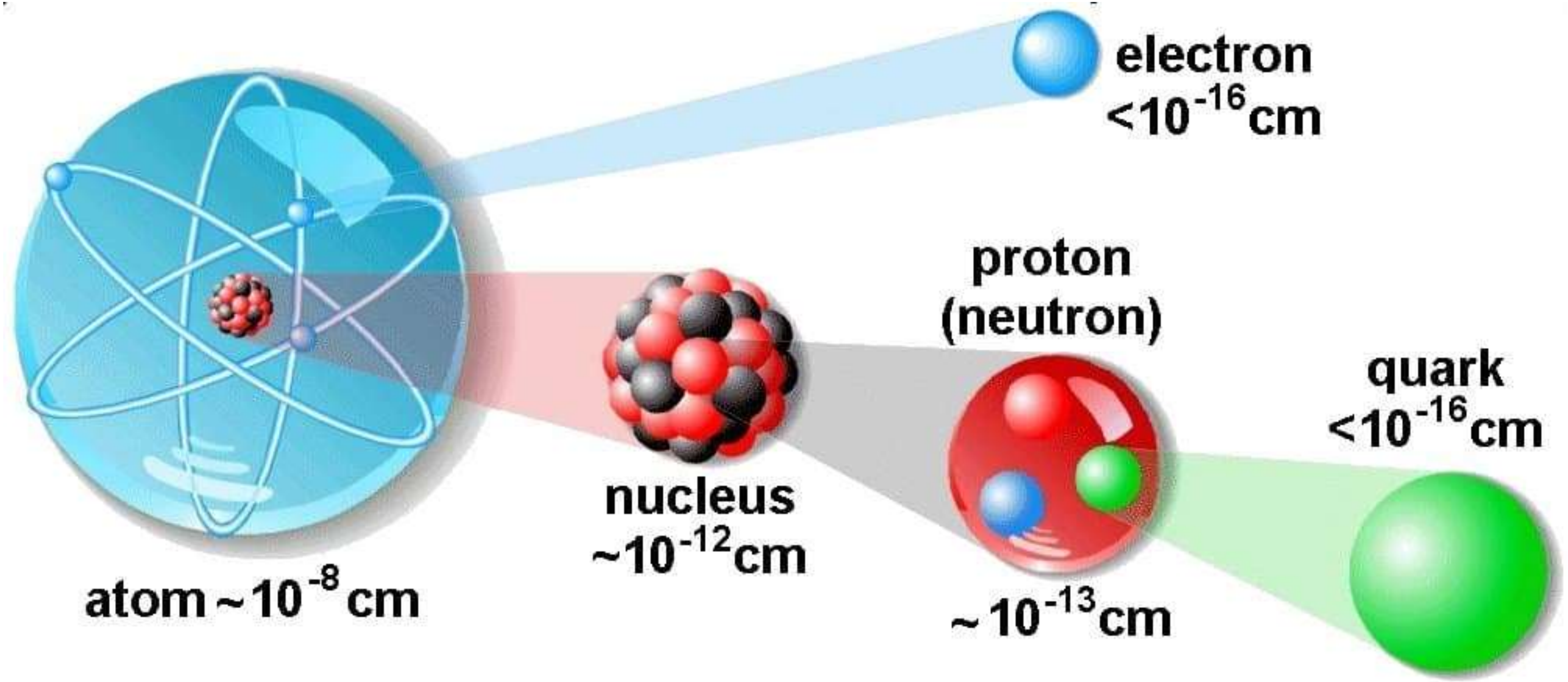
Nuclear Physics is an important branch of science that facilitates the understanding of nuclear reactions. Thus, over here reactions, radioactive decay, nuclei fusion, and fissions can be studied in a very detailed manner.

It includes the study of ,

1. The general properties of nucleus.
2. The particles contained in the nucleus.
3. The interaction between these particles.
4. Radio activity and nuclear reactions.
5. Practical applications of nuclear phenomena.

MATTER
from molecule to quark




electron
<10^-16 cm
proton
(neutron)
quark
<10^-16 cm
nucleus
~10^-12 cm
atom ~10^-8 cm
~10^-13 cm

NUCLEUS

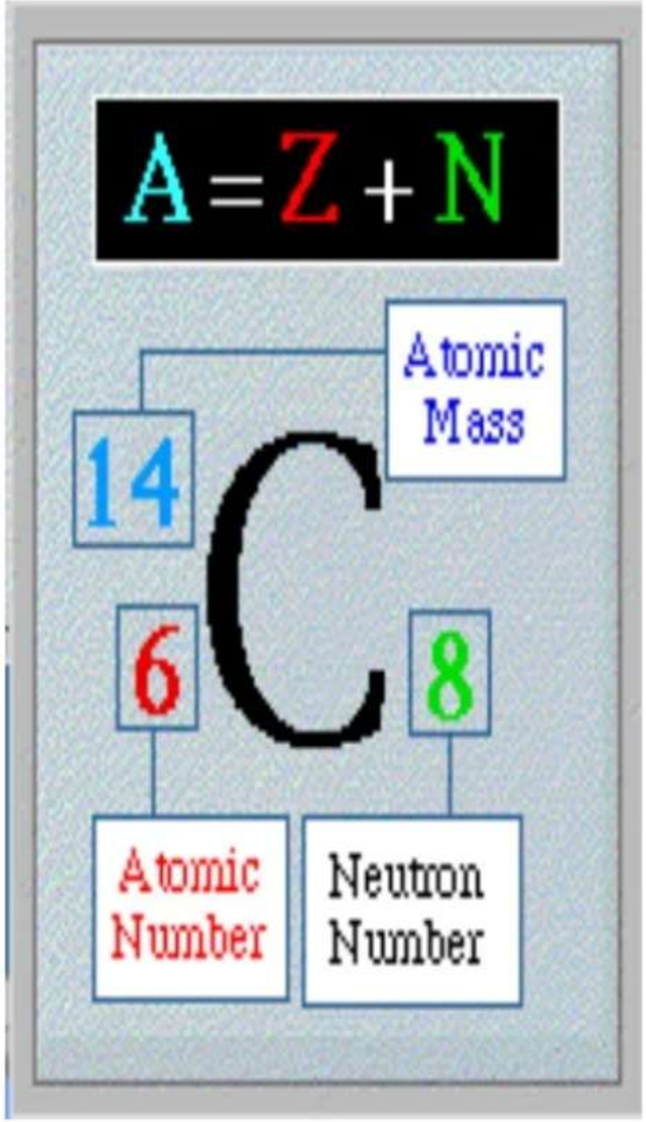
- MASS NUMBER(**A**): total number of nucleon. $A=Z(protons)+N(neutrons)$.

- ATOMIC NUMBER(**Z**): number of protons.

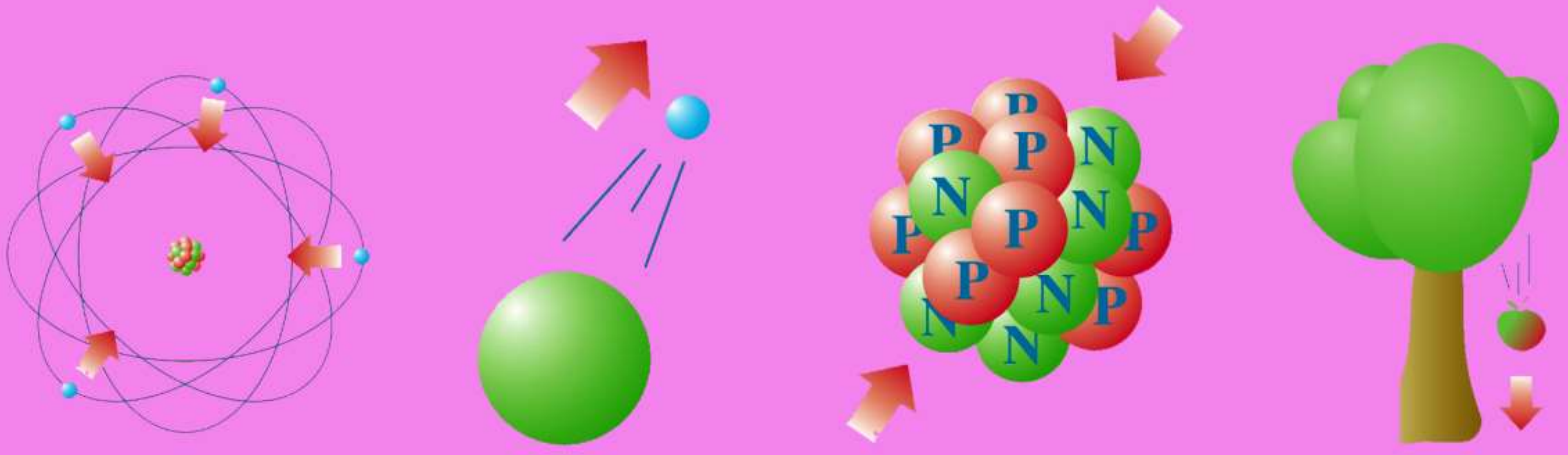
- NEUTRON NUMBER N: number of neutrons.

- RADIUS: $\mathbf{r=r_0A^{1/3}}$, $r_0=1.25\times10^{-15}$ m.

- MASS **M=AU**, $u=1.66\times10^{-27}$kg



FUNDAMENTAL FORCES OF NATURE



Electro-magnetism

Weak Interaction

Strong Interaction

Gravitation

Four Fundamental Forces

<u>Gravity</u>

- Attractive only
- Property of Mass
- Light responds to gravity
- Can be thought of as a reshaping of space-time

<u>Electro-Magnetism</u>

- Interaction of charged particles
- Responsible for ALL surface interactions of matter
- Responsible for molecular bonding

<u>Strong Nuclear Force</u>

- Holds the Protons and Neutrons together in the nucleus of the atom

<u>Weak Nuclear Force</u>

- Responsible for nuclear radiation & fission

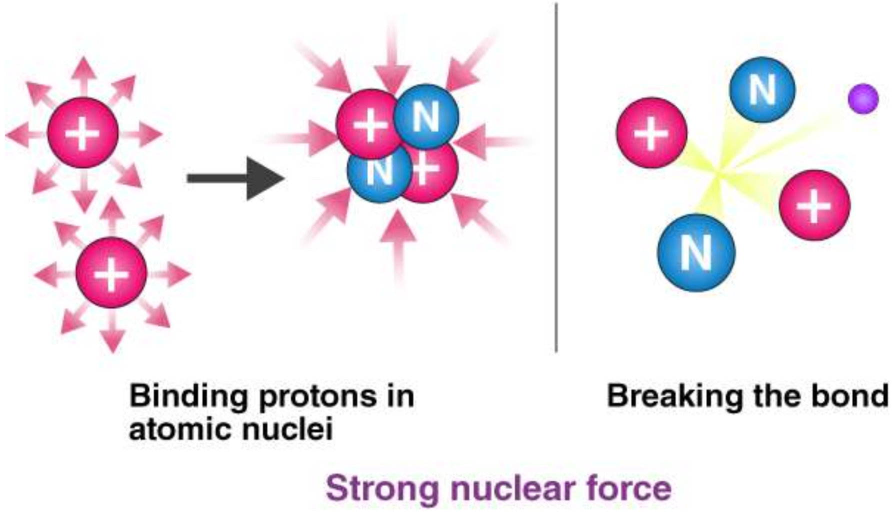
Fundamental Forces

Force	Illustration	Strength	Range (m)	Particle
Strong	 Force which holds nucleus togeher	1	10^{-15} (diameter of a medium sized nucleus)	gluons, π(nucleons)
Electro-magnetic		$\frac{1}{137}$	Infinite	photon mass = 0 spin = 1
Weak	 neutrino interaction induces beta decay	10^{-6}	10^{-18} (0.1% of the diameter of a proton)	Intermediate vector bosons W^+, W^-, Z_0, mass > 80 GeV spin =1
Gravity		6×10^{-39}	Infinite	graviton ? mass = 0 spin = 2

THE NUCLEAR FORCE :-

The force that binds together protons and neutrons inside the nucleus is called the Nuclear Force.

This force can exist between protons and protons, neutrons and protons or neutrons and neutrons. This force is what holds the nucleus together.



Some characteristics of the nuclear force are:

1. It does not depend on the charge.
2. It is very short-range.
3. It is much stronger than the electric force.
4. It is a saturated force.
5. It favors the formation of pairs of nucleons with opposite spins.

BINDING ENERGY :-

Nuclear binding energy is the energy required to break up the nucleus into its separate nucleons OR this can be expressed as the energy released when the nucleus is formed from separate nucleons.

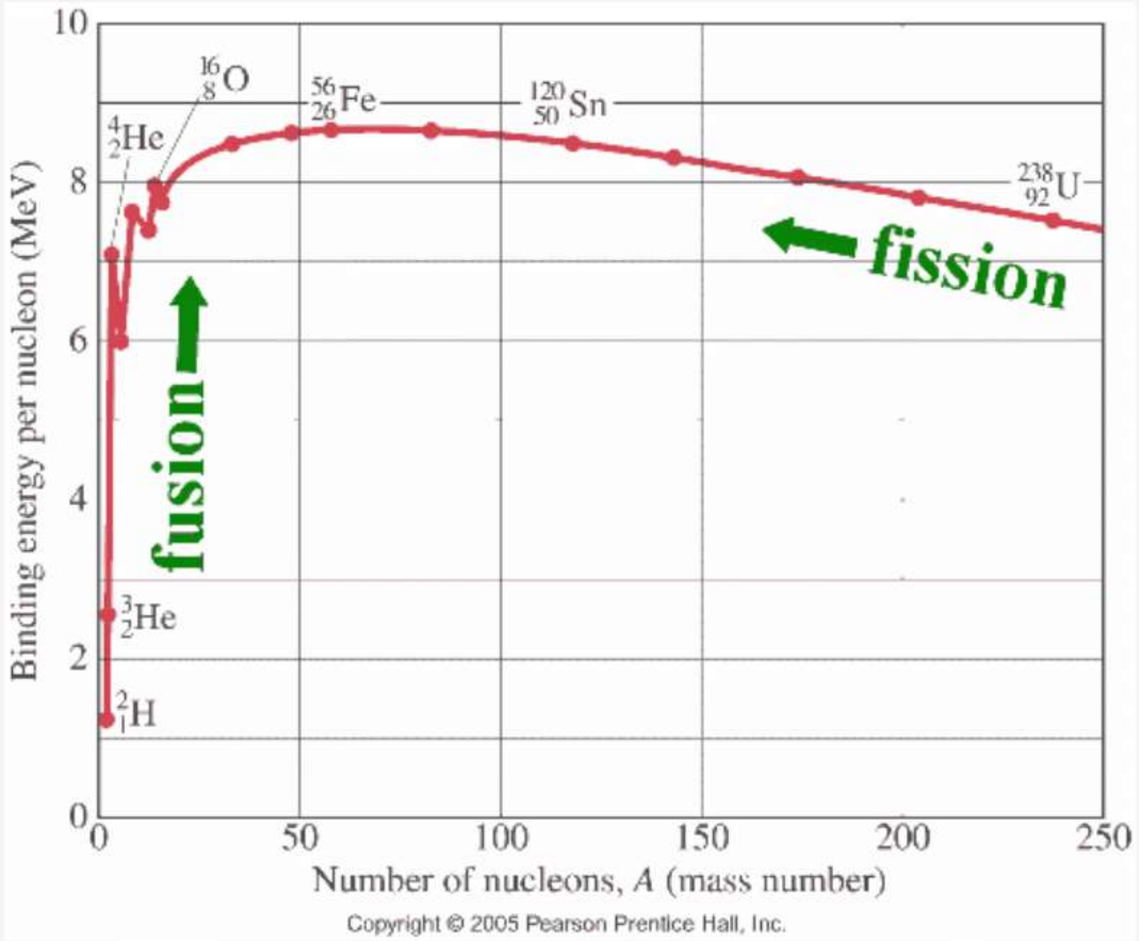
BINDING ENERGY PER NUCLEON :-

If we know the total binding energy of a nucleus, and the number of nucleons, we can work out the binding energy per nucleon, which is the average energy needed to remove each nucleon. You have to be very careful about distinguishing between 'binding energy' and 'binding energy per nucleon.!

Note:- THE HIGHER THE BINDING ENERGY PER NUCLEON, THE MORE STABLE THE NUCLEUS IS.

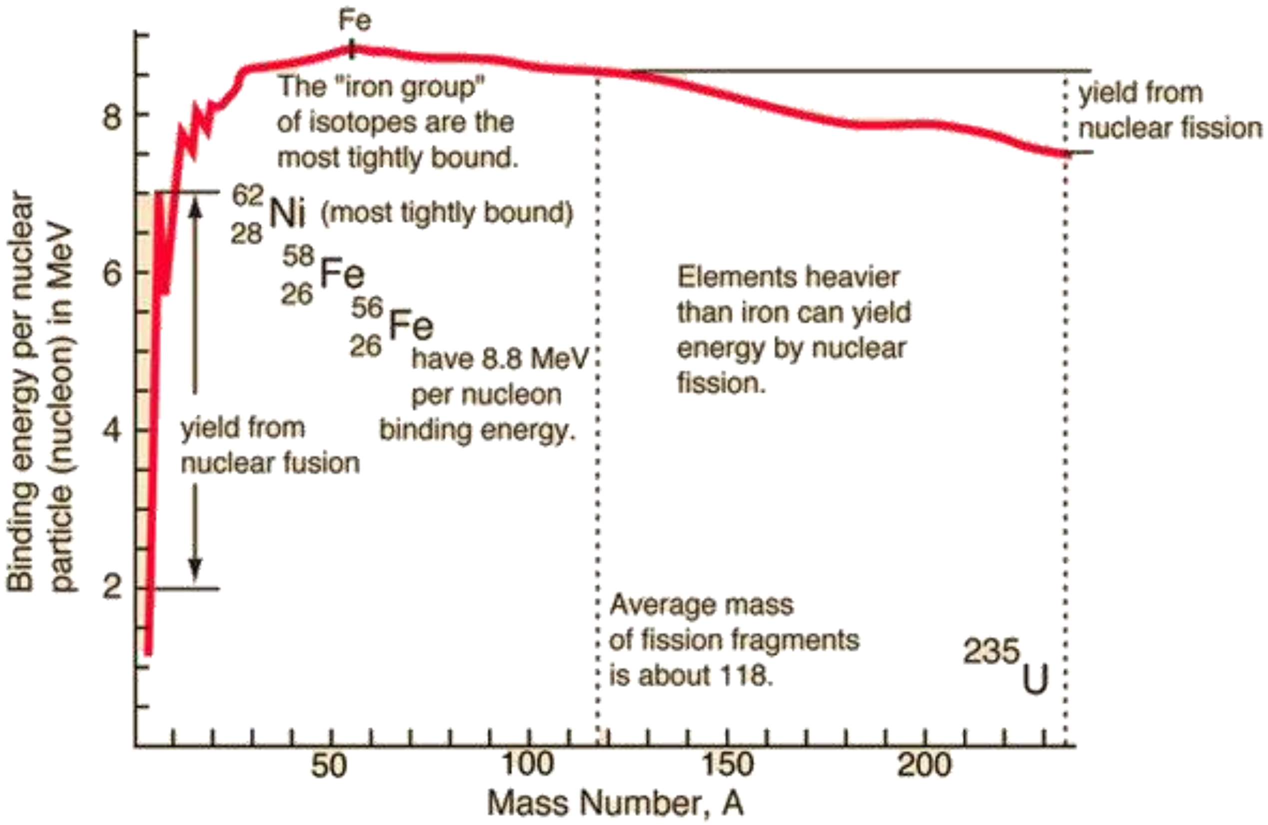
Binding Energy Curve:- Binding Energy per Nucleon vs. Mass Number

- This Curve illustrates that as the atomic mass number increases, the binding energy per nucleon decreases for A > 60.
- The BE /A curve reaches a maximum value of 8.79 MeV at A = 56 and decreases to about 7.6 MeV for A = 238.
- The general shape of the BE v/s A curve can be explained using the general properties of nuclear forces. The nucleus is held together by very short-range attractive forces that exist between nucleons. On the other hand, the nucleus is being forced apart by long-range repulsive electrostatic (coulomb) forces that exist between all the protons in the nucleus.
- As the atomic number and the atomic mass number increase, the repulsive electrostatic forces within the nucleus increase due to the greater number of protons in the heavy elements. To overcome this increased repulsion, the proportion of neutrons in the nucleus must increase to maintain stability.



- This increase in the neutron-to-proton ratio only partially compensates for the growing proton-proton repulsive force in the heavier, naturally occurring elements. Because the repulsive forces are increasing, less energy must be supplied, on the average, to remove a nucleon from the nucleus. The BE/A has decreased. The BE/A of a nucleus is an indication of its degree of stability. Generally, the more stable nuclides have higher BE/A than the less stable ones.

The increase in the BE/A as the atomic mass number decreases from 260 to 60 is the primary reason for the energy liberation in the fission process. In addition, the increase in the BE/A as the atomic mass number increases from 1 to 60 is the reason for the energy liberation in the fusion process, which is the opposite reaction of fission.

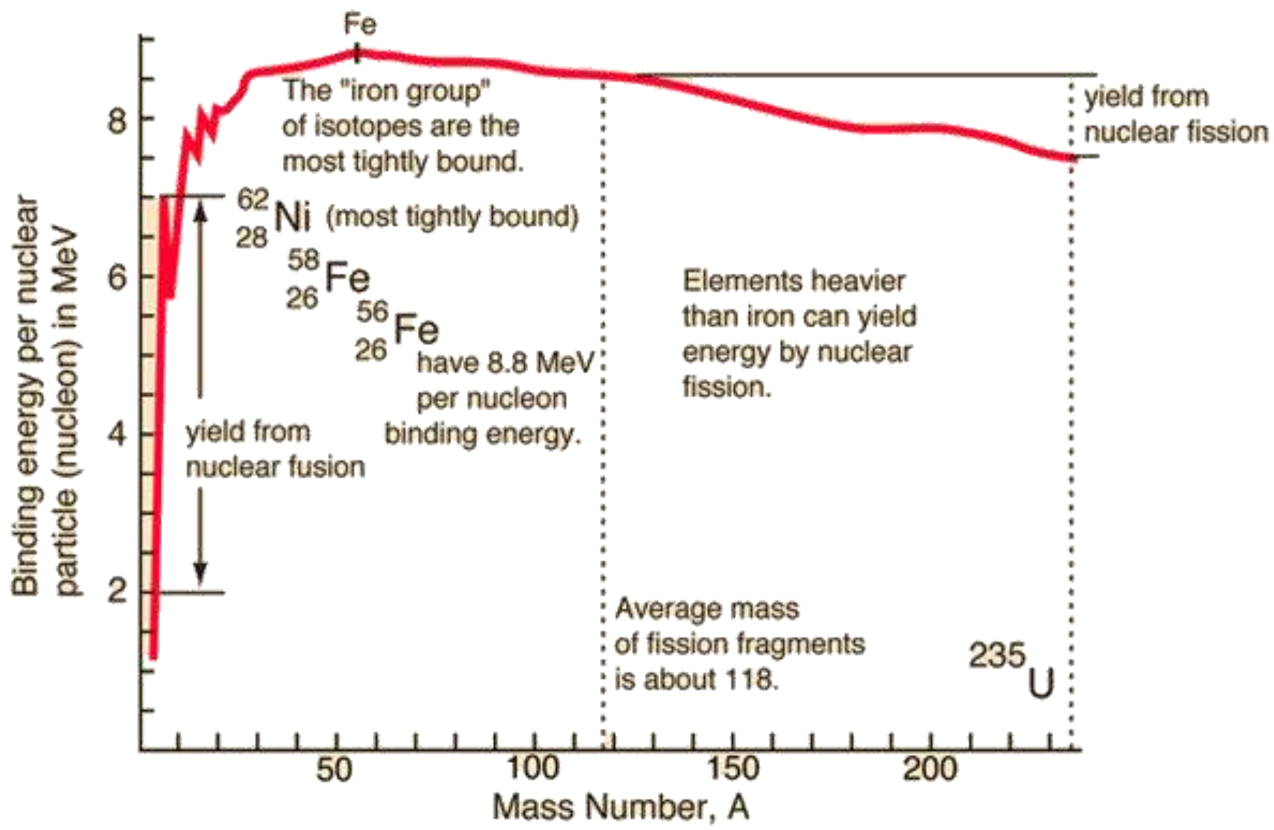


The heaviest nuclei require only a small distortion from a spherical shape (small energy addition) for the relatively large coulomb forces forcing the two halves of the nucleus apart to overcome the attractive nuclear forces holding the two halves together. Consequently, the heaviest nuclei are easily fissionable compared to lighter nuclei.

Binding Energy Curve:- Binding Energy per Nucleon vs. Mass Number

Nuclei larger than iron undergo nuclear fission

If we look at large nuclei (much greater than iron), we find that the further to the right on the graph we go (going to greater nucleon numbers) the less stable the nuclei. This is because the binding energy **per nucleon** is getting less. These nuclei undergo **fission** and split to produce products with higher binding energy per nucleaon values - more stable products.



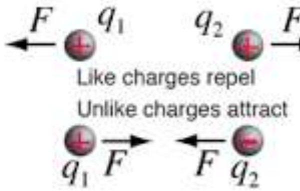
Nuclei smaller than iron undergo fusion

If we look at nuclei of much smaller mass than iron we find they have a lower binding energy per nucleon. Therefore when these fuse to produce a heavier nucleus it is more stable - **fusion** can be shown to be energetically viable from the above graph. The product nucleus has a higher binding energy pernucleon than the two that fuse to form it. It is therefore more stable than its constituents.

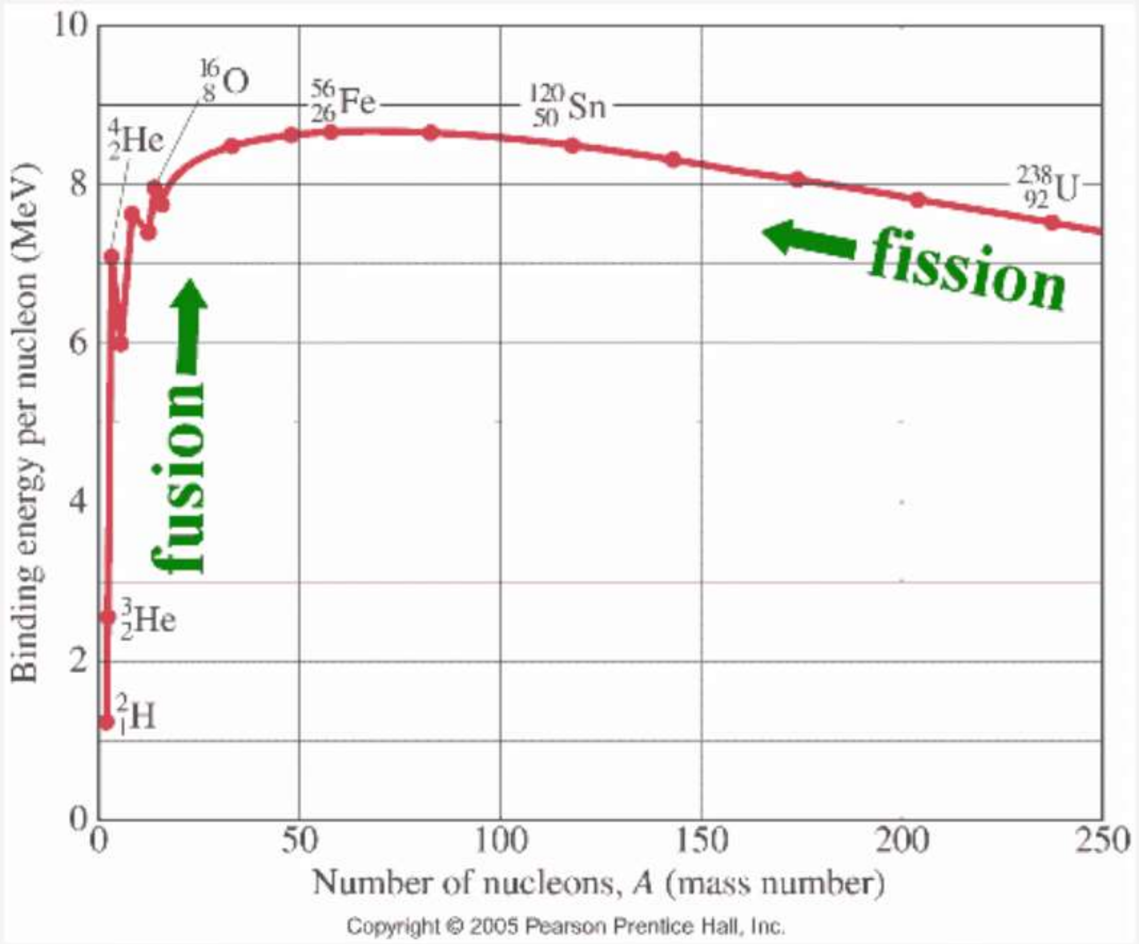
Question:-Why Binding energies per nucleon increase sharply as Mass Number (A) increases?

Ans:- The binding energy per nucleus is determined by dividing the binding energy of the nucleus by the number of nucleons it contains. Elements with intermediate atomic masses have the greatest binding energies per nucleon and are therefore the most stable.

The decrease in binding energy per nucleon for heavy nuclei (A>100) due to an increase in coulomb repulsion between the protons inside the nucleus.



$$F = \frac{kq_1q_2}{r^2} = \frac{q_1q_2}{4\pi\varepsilon_0 r^2}$$ *Coulomb's Law*



Question: -What is the reason for spikes in this Binding Energy Curve?

Ans:-

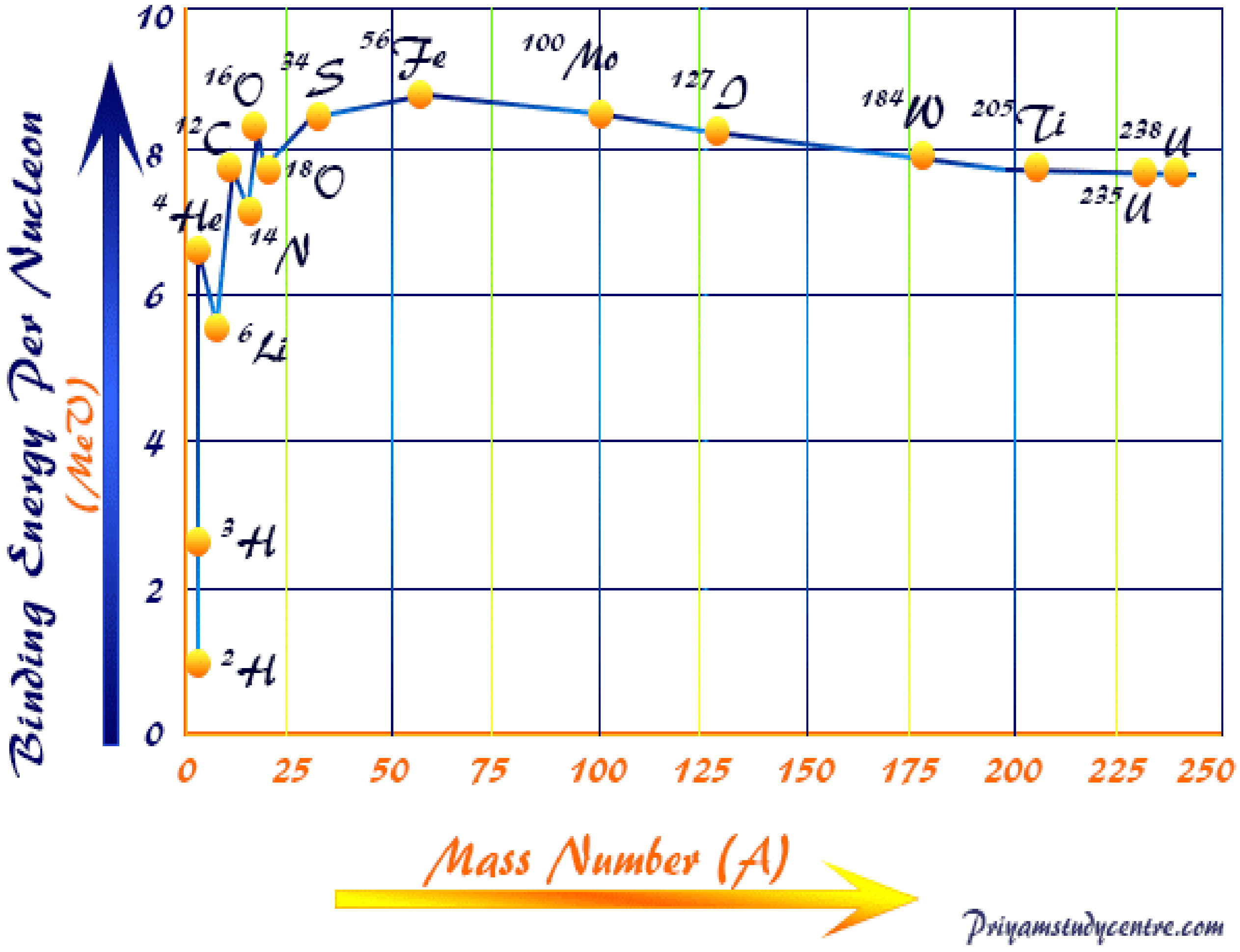
There are two effects that lead to the presence of the small, jagged peaks and valleys in the binding energy per nucleon. (**The main shape of the curve is given by the semi-empirical mass formula, derived from the liquid drop model of the nucleus.** The model has positive binding energy proportional to the number of adjacent nucleon-nucleon pairs, a Coulomb repulsion term related to the square of the number of protons, and a term related to the proton-neutron imbalance.) **On top of this, producing the zigzags, there are pairing effects and magic number effects.**

(i) The pairing effects come from the fact that the bound nucleons have slightly lower energies when they are correlated in proton-proton or neutron-neutron pairs. That tends to make the binding energy per nucleon for an odd-odd nucleus like ^{10}B less than the odd-even ^{11}B.

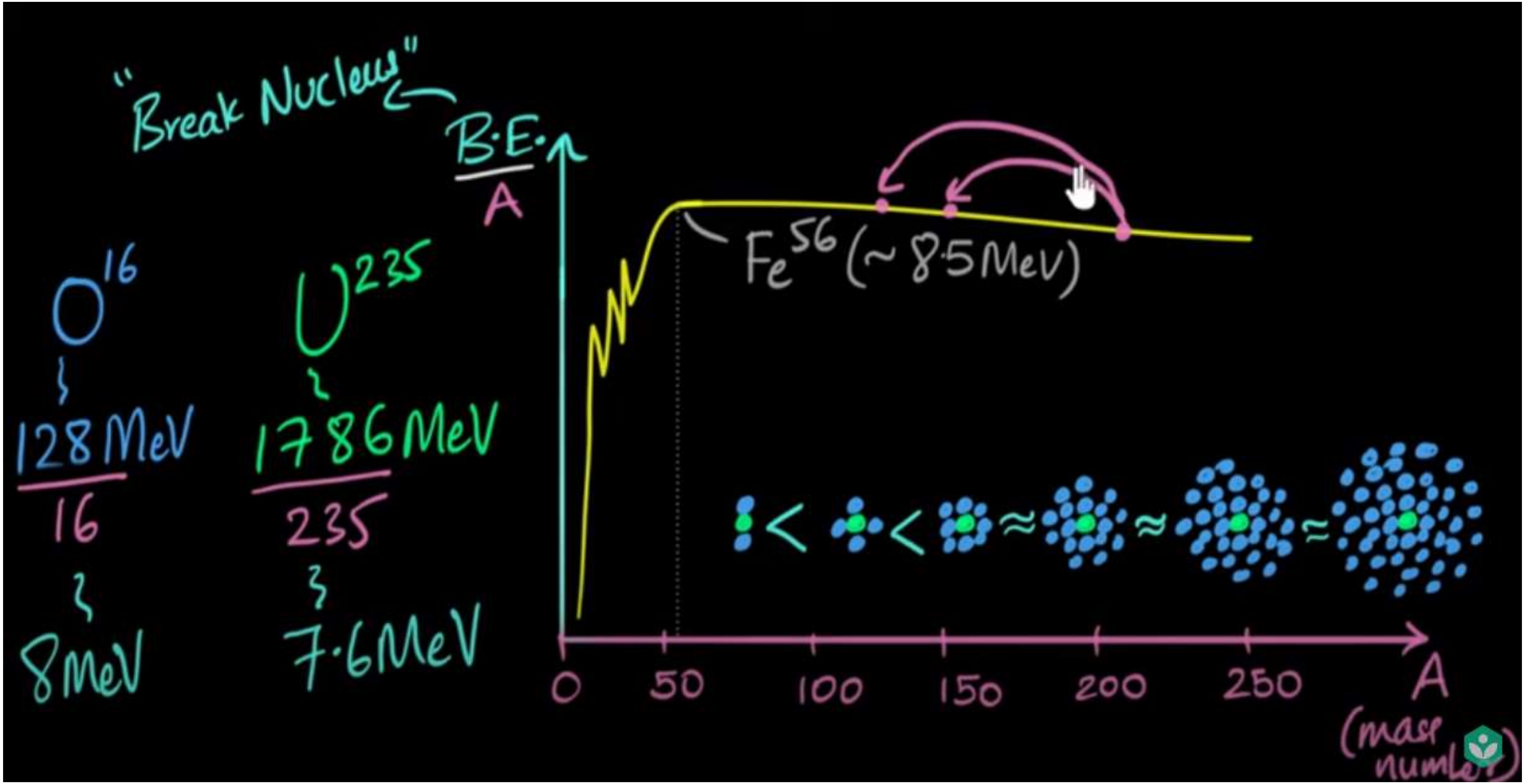
OR **17O is less tightly bound than 16O and 18O on either side of it, since 16O and 18O are both even-even.**

(ii) The other effect is due to the presence of magic numbers, which are related to filled nuclear orbitals. Just like atomic electrons are most stable when they form a filled outer shell, nuclei are most stable when the protons and/or neutrons fill certain nuclear shells.

For example, 4He is much more tightly bound than 3H or 3He, since the 4He has two protons and two neutrons, with each pair filling a 1S shell. Another (double) magic nucleus is 16O, with the eight protons and neutrons each filling the 11S and 11P shells. (The notation for shells differs a bit here from that used with electrons. Nuclear shells are denoted by nS, nP, etc., where n starts separately at one for each value of the angular momentum.) The 18O nucleus has to have its two extra neutrons shunted into the higher-energy 2S shell, lowering the binding energy per nucleon. Another magic number occurs at 10, which is why 20Ca is especially stable; the ten protons and ten neutrons fill-up the 1S, 1P, and 2S shells. (The pattern of magic numbers gets a bit more complicated than this, because of the strong spin-orbit coupling in the nucleus, but this is a reasonable picture of the general behavior.)



Summary:- Due to equal protons and neutrons they have extra stability w.r.t. their average curve and extra stability implies more binding energy hence a peak.



"Break Nucleus"
B.E.
A
O^{16}
128 MeV
16
8MeV
U^{235}
1786 MeV
235
7.6MeV
Fe^{56} (~8.5MeV)
0
50
100
150
200
250
A
(mass number)

A HISTORY OF THE ATOM: THEORIES AND MODELS

How have our ideas about atoms changed over the years? This graphic looks at atomic models and how they developed.

SOLID SPHERE MODEL

JOHN DALTON

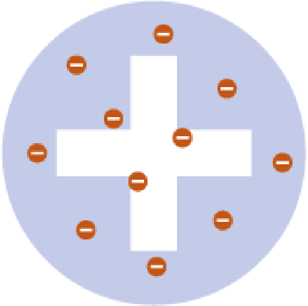
1803

Dalton drew upon the Ancient Greek idea of atoms (the word 'atom' comes from the Greek '*atomos*' meaning indivisible). His theory stated that atoms are indivisible, those of a given element are identical, and compounds are combinations of different types of atoms.

+ RECOGNISED ATOMS OF A PARTICULAR ELEMENT DIFFER FROM OTHER ELEMENTS

− ATOMS AREN'T INDIVISIBLE - THEY'RE COMPOSED FROM SUBATOMIC PARTICLES

PLUM PUDDING MODEL



J.J. THOMSON

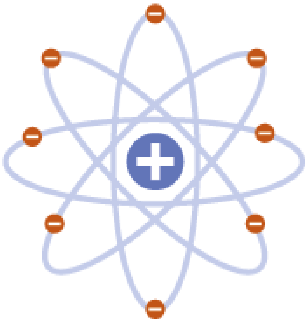
1904

Thomson discovered electrons (which he called 'corpuscles') in atoms in 1897, for which he won a Nobel Prize. He subsequently produced the 'plum pudding' model of the atom. It shows the atom as composed of electrons scattered throughout a spherical cloud of positive charge.

+ RECOGNISED ELECTRONS AS COMPONENTS OF ATOMS

− NO NUCLEUS; DIDN'T EXPLAIN LATER EXPERIMENTAL OBSERVATIONS

NUCLEAR MODEL



ERNEST RUTHERFORD

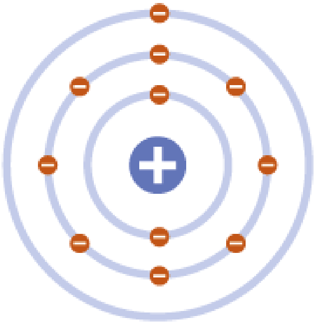
1911

Rutherford fired positively charged alpha particles at a thin sheet of gold foil. Most passed through with little deflection, but some deflected at large angles. This was only possible if the atom was mostly empty space, with the positive charge concentrated in the centre: the nucleus.

+ REALISED POSITIVE CHARGE WAS LOCALISED IN THE NUCLEUS OF AN ATOM

− DID NOT EXPLAIN WHY ELECTRONS REMAIN IN ORBIT AROUND THE NUCLEUS

PLANETARY MODEL



NIELS BOHR

1913

Bohr modified Rutherford's model of the atom by stating that electrons moved around the nucleus in orbits of fixed sizes and energies. Electron energy in this model was quantised; electrons could not occupy values of energy between the fixed energy levels.

+ PROPOSED STABLE ELECTRON ORBITS; EXPLAINED THE EMISSION SPECTRA OF SOME ELEMENTS

− MOVING ELECTRONS SHOULD EMIT ENERGY AND COLLAPSE INTO THE NUCLEUS; MODEL DID NOT WORK WELL FOR HEAVIER ATOMS

QUANTUM MODEL

ERWIN SCHRÖDINGER

1926

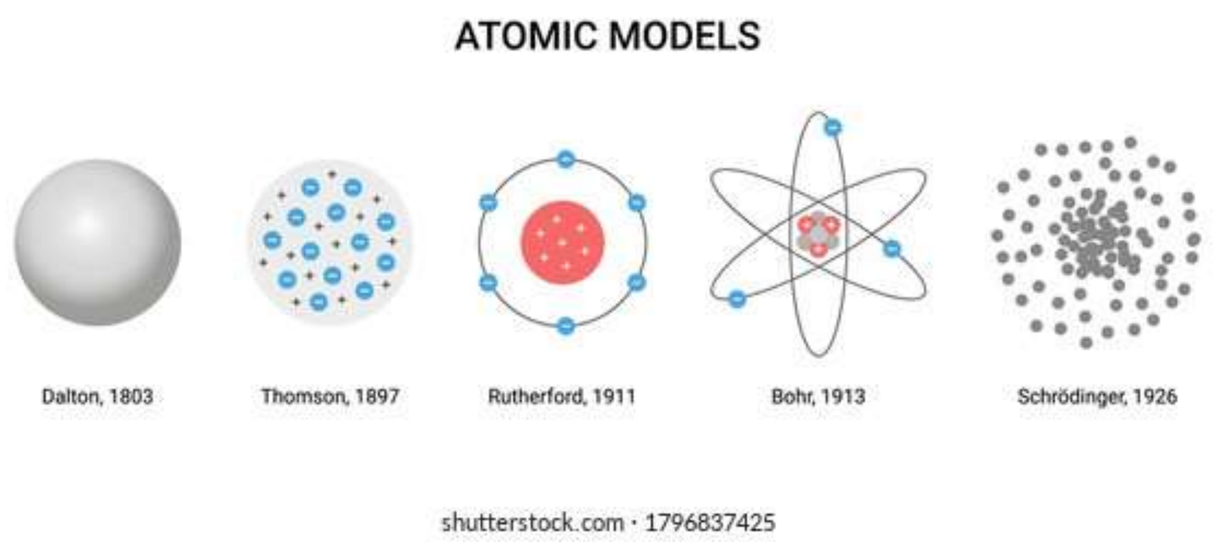
Schrödinger stated that electrons do not move in set paths around the nucleus, but in waves. It is impossible to know the exact location of the electrons; instead, we have 'clouds of probability' called orbitals, in which we are more likely to find an electron.

+ SHOWS ELECTRONS DON'T MOVE AROUND THE NUCLEUS IN ORBITS, BUT IN CLOUDS WHERE THEIR POSITION IS UNCERTAIN

+ STILL WIDELY ACCEPTED AS THE MOST ACCURATE MODEL OF THE ATOM

Atomic model:-

The atomic model which describes rotation of electrons in different orbitals around the positively charged nucleus is called the nuclear model of atom.



Nuclear models can be classified into two main groups.

In those of the first group, called independent-particle models, the main assumption is that little or no interaction occurs between the individual particles that constitute nuclei; each proton and neutron moves in its own orbit and behaves as if the other nuclear particles were passive participants. The shell nuclear model (*q.v.*) and its variations fall into this group.

In a second group, called strong-interaction, or statistical models, the main assumption is that the protons and neutrons are mutually coupled to each other and behave cooperatively in a way that reflects the short-ranged strong nuclear force between them. The liquid-drop model and compound-nucleus model (*qq.v.*) are examples of this group.

Other nuclear models incorporate aspects of both groups, such as the collective model (*q.v.*), which is a combination of the shell model and the liquid-drop model.

NUCLEAR MODELS

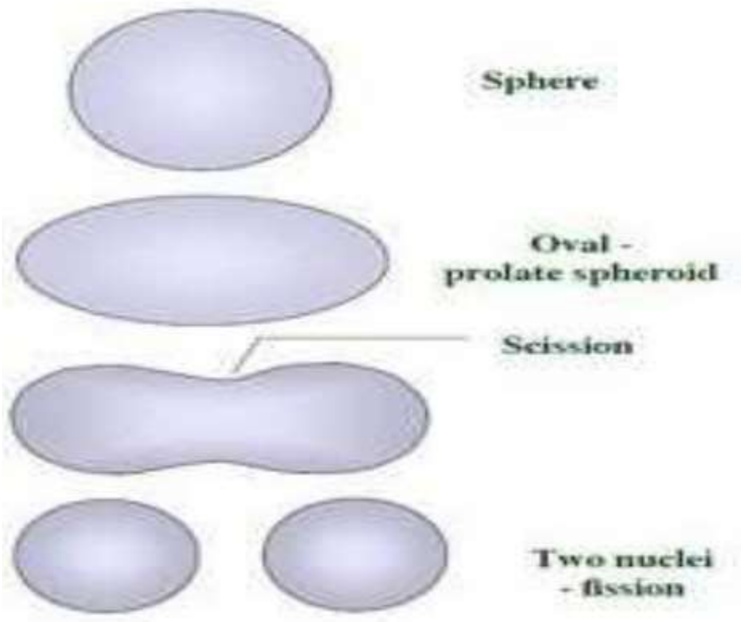
- ❑Shell model
- ❑Collective model
- ❑Fermi gas model
- ❑Liquid drop model
- ❑Compound nuclear model
- ❑Direct interaction model
- ❑Optical model

The water-drop model or Liquid drop model (LDM)

(It has not individual particle model with nucleons in discrete energy states):-

The liquid drop model was first proposed by George Gamow and further developed by Niels Bohr and John Wheeler.

The Liquid Drop Model treats the nucleus as a liquid. Nuclear properties, such as the binding energy, are described in terms of volume energy, surface energy, compressibility, etc, parameters that are usually associated with a liquid. This model has been successful in describing how a nucleus can deform and undergo fission.



Liquid Drop Model of Nucleus

(Similarities between liquid drop and nucleus)

Liquid drop	Nucleus
Cohesive forces between liquid molecules	Strong force between nucleons
The density of liquid in liquid drop is constant	Nuclear density is constant and is independent of nuclear volume
When heat is supplied to a drop, it evaporates	In radioactive decay some nucleons have enough energy to jump out of the nucleus
Liquid drop has surface tension	Nucleus has surface energy proportional to $A^{2/3}$
When a liquid drop is excited it splits into two or more drops	During fission, an incident particle excites a heavy nucleus and it breaks into fractions

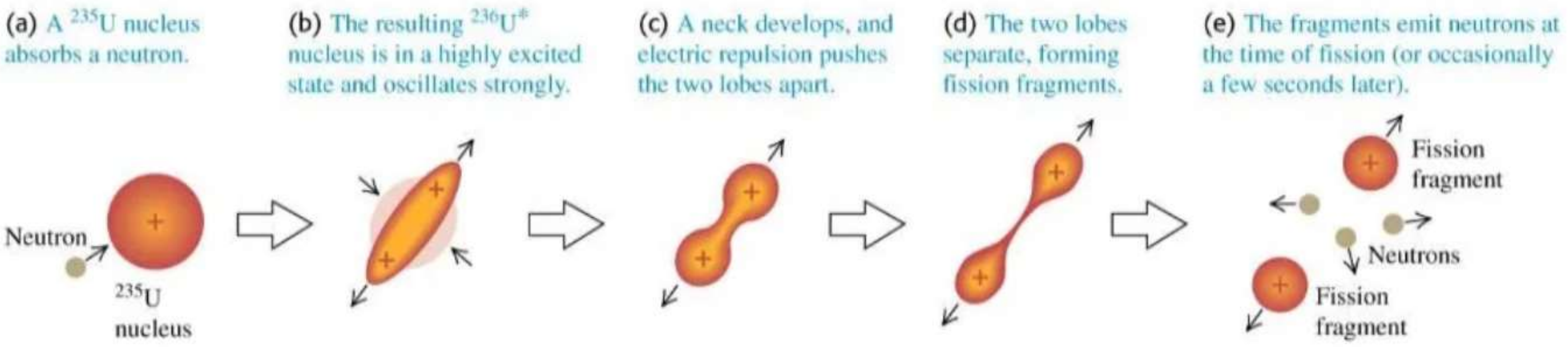
Liquid Drop Model of Nucleus

(Similarities between liquid drop and nucleus)

Liquid drop	Nucleus
When two liquid drops are brought closer to each other they combine to form a bigger liquid drop	Two lighter nuclei combine to form a compound nucleus
Liquid drop is spherical in shape due to spherical symmetry of surface tension	In stable, state nucleus is spherical in shape

Assumptions :-

1. The nuclei of all elements are considered to behave like a liquid drop of incompressible liquid of very high density.
2. In an equilibrium state the nuclei of atoms remain spherically symmetric under the action of strong attractive nuclear forces just like the drop of a liquid which is spherical due to surface tension.
3. The density of a nucleus is independent of its size just like the density of liquid which is also independent of its size.
4. The nucleons of the nucleus move about within a spherical enclosure called the nuclear potential barrier just like the movement of the molecules of a liquid within a spherical drop of liquid.
5. The binding energy per nucleon of a nucleus is constant just like the latent heat of vaporization of a liquid.
6. Nuclear force is independent of charge and spin of the nucleons.



Nuclear Shell Model (An individual particle model with nucleons in discrete energy states):-

The nuclear shell model is a model of the atomic nucleus. It uses the Pauli exclusion principle to explain the nucleus structure in terms of energy levels. In order to study the complete nucleus structure, various nuclear shell models were proposed.

The first shell model was introduced in the year 1932 by Dmitry Ivanenko and later on developed by various physicists – Maria Goeppert Mayer, Eugene Paul Wigner, and J.Hans.D. Jensen in the year 1949.

Nuclides with a magic number of neutrons are observed to have a relatively low probability of absorbing an extra neutron, i.e. they have lowest of absorption cross sections for neutrons (neutron-capture cross sections). To explain such nuclear systematics and the internal structure of the nucleus, a shell model of the nucleus has been developed. **This model uses Schrodinger's wave equation or quantum mechanics to describe the energetics of the nucleons in a nucleus in a manner analogous to that used to describe the discrete energy states of electrons around the nucleus.**

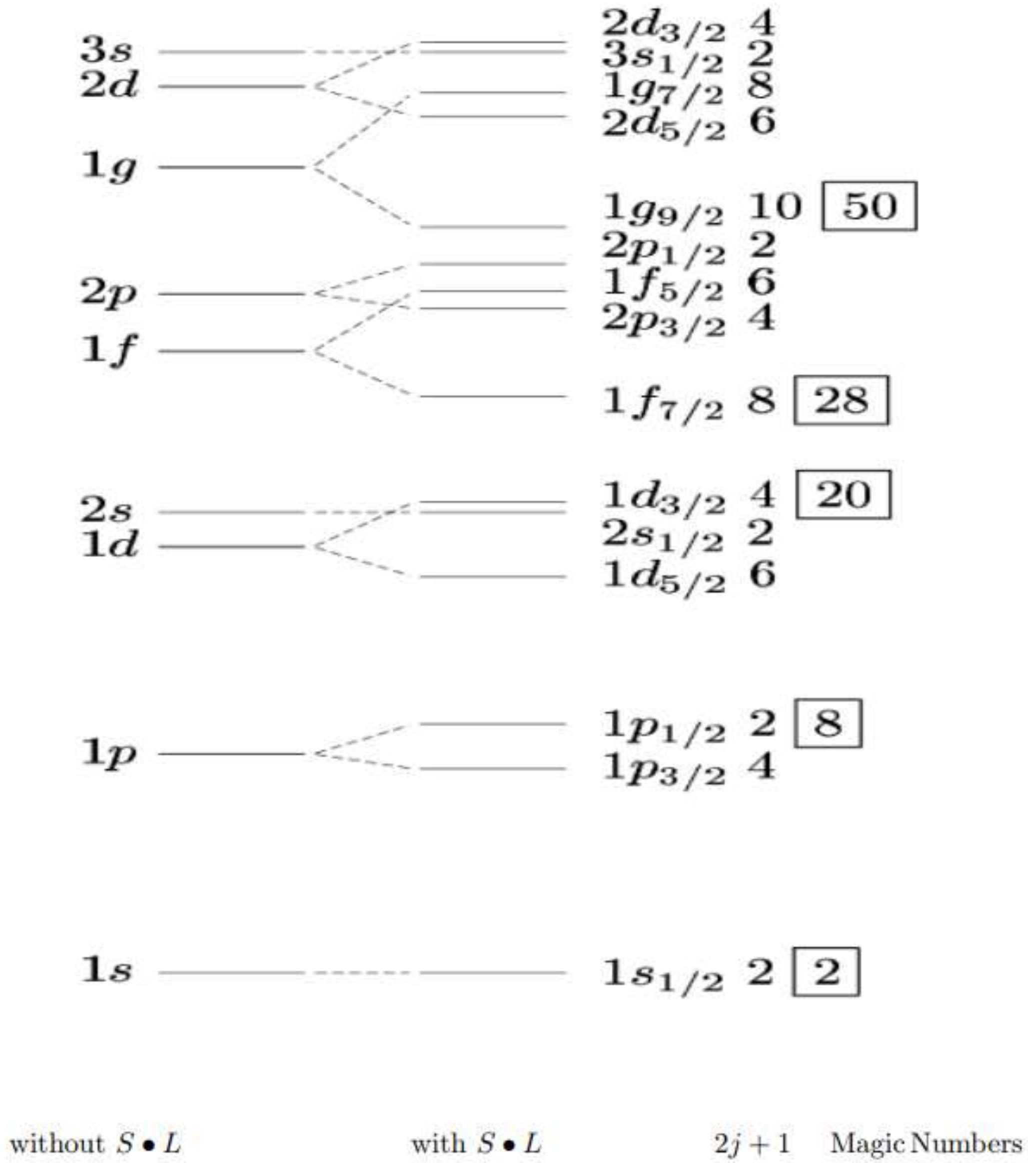
Nuclear Shell Model Assumes:

1. Each nucleon moves independently in the nucleus uninfluenced by the motion of the other nucleons.
2. Each nucleon moves in a potential well which is constant from the center of the nucleus to its edge where it increases rapidly by several tens of MeV. When the model's quantum-mechanical wave equation is solved numerically, the nucleons are found to distribute themselves into a number of energy levels.
3. There is a set of energy levels for protons and an independent set of levels for neutrons. Filled shells are indicated by large gaps between adjacent energy levels and are computed to occur at the experimentally observed values of 2, 8, 20, 28, 50, 82, and 126 neutrons or protons. Such closed shells are analogous to the closed shells of orbital electrons.

However, the shell model has been useful to obtain such results that predicts the magic numbers and particularly useful in predicting several properties of the nucleus, including (1) the total angular momentum of a nucleus, (2) characteristics of isomeric transitions, which are governed by large changes in nuclear angular momentum, (3) the characteristics of beta decay and gamma decay, and (4) the magnetic moments of nuclei.

Table (3-1): Arrangement of the nuclear shells and its distributions.

ℓ / n	0	1	2	3	4	5	6	shell	No. of nucleons
1	1s							s	2
2		1p						p	6
3	2s		1d					d	10
4		2p		1f				f	14
5	3s		2d		1g			g	18
6		3p		2f		1h		h	22
7	4s		3d		2g		1i	i	26



Where S . L is Spin-orbit interaction

The nuclear energy levels going to be filled up by the nucleons in various shells

Achievements of the nuclear shell model :

1. It Explains nuclear spin and parity.
2. It also explains magnetic moments for lighter nuclei.
3. It also speaks about excited states of nuclei.

Failures of this model:

1. For heavier nuclei it fails to predict magnetic dipole moments or the spectra of excited states very well.
2. The Shell Model also does not give predictions of electric quadrupole moments of the nuclei.

Applications of Nuclear Physics

Nuclear Physics has a wide range of applications in today's world. Some of its major areas of contributions include

1. Carbon dating of the specimen in Archeology and Geology.
2. Nuclear Medicine for usability in drug administration and radiation therapy
3. In the generation of usable energy
4. Nuclear Fission reaction

What are the disadvantages of Nuclear energy Application?

Nuclear Energy is advantageous on numerous levels. Moreover, there are certain aspects of the energy emission which Nuclear Energy creates that have a negative impact on the environment. These include:

1. Hazardous and toxic radioactive waste causing deadly altercations on the living body
2. Extremely high maintenance and operation cost
3. Radioactive remnants that have a decay time of thousands of ears causing extreme environmental issues.
4. Uranium, the core raw material needed for Nuclear reaction, is depleting from the earth's surface at an alarming rate. Moreover, we cannot replenish it, once we consume it completely.

Nuclear Fission and Nuclear Fusion